The Exponential Wall:
Complete Bessel Spectrum and Resolvent

HCS-C398 theorem and reproducibility package

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Abstract

The positive exponential potential on the Dirichlet half-line has a complete, simple Bessel-order spectrum and a natural self-adjoint quantization of a classical bouncing orbit family. We derive its full resolvent and the boundary derivative that normalizes every eigenfunction. Pólya and Lagarias own the classical source mechanism; the number-theoretic oscillation input is explicitly cited, not reproved or fitted from data. A real logarithmic Weyl law is not a target spectral identification.

Keywords: exponential potential; Bessel order zero; spectral determinant; Weyl remainder; heat trace; counting obstruction.

中文摘要

本文研究半直线上的正指数势阱，给出完整单重离散谱、预解式及全部特征函数的归一化。 经典势阱理论与外部数论振荡定理均明确归属，不使用目标零点表拟合， 也不将自然自伴算子直接宣称为目标算子。
关键词：指数势；贝塞尔阶零点；谱行列式；计数余项；热迹；匹配障碍。

1 Frozen half-line source and classical clock

For every $a > 0$, let H_a be the self-adjoint operator associated with

$$q_a[u] = \int_0^\infty (|u'|^2 + a^2 e^{2x}|u|^2)\,dx, \quad D(q_a) = H_0^1(0,\infty) \cap L^2(e^{2x}dx).$$

Its domain consists of u, u' locally absolutely continuous, $u(0) = 0$, $u \in L^2$ and $-u'' + a^2 e^{2x} u \in L^2$. Infinity is limit point. The kinetic coefficient and $\hbar$ are one; E denotes energy and $k = \sqrt{E}$ positive frequency. Pólya's order-zero phenomenon is classical, as documented by Lagarias [1]; that paper treats the more general right-half-line Morse potential. Our project's full-line Morse system instead has finitely many bound levels and a continuum. Changing the domain and confinement class here is mathematically essential. The source results below are reconstructed, not asserted to have new literature priority.

The classical Hamiltonian is $p^2 + a^2 e^{2x}$ with specular reflection at $x = 0$. Every $E = k^2 > a^2$ has one bouncing orbit, turning at $x_* = \log(k/a)$. Its full action and physical period are

$$J(E) = 2[k \operatorname{arcosh}(k/a) - \sqrt{k^2 - a^2}], \tag{1}$$
$$T(E) = J'(E) = \operatorname{arcosh}(k/a)/k. \tag{2}$$

Indeed substitute $y = ae^x/k$ in $2\int_0^{x_*} \sqrt{E - a^2 e^{2x}}dx$. The endpoint integrand vanishes, so differentiation gives the period; $\dot{x} = 2p$ fixes its factor. Reversal is $(x, p) \mapsto (x, -p)$. Below a^2 the energy shell is empty, and equality is a degenerate wall threshold, not a nontrivial cycle. This continuous energy family is not a rational-prime-indexed list.

2 Complete spectrum, Green kernel and normalization

Theorem 1. *H_a has compact resolvent and a complete orthogonal basis of simple eigenfunctions. Its spectrum is exactly*

$$a^2 < E_1 < E_2 < \cdots \longrightarrow \infty, \qquad K_{i\sqrt{E_n}}(a) = 0,$$

and $y_n(x) = K_{i\sqrt{E_n}}(ae^x)$ spans the corresponding eigenspace. There are no other energy zeros of $K_{\sqrt{-E}}(a)$.

Proof. The form is dense, closed and bounded below by a^2. On its unit ball the squared L^2 tail beyond R is at most $a^{-2}e^{-2R}$. Compact interval H^1 embedding and this tail bound give compact form embedding and compact resolvent. The spectral theorem then gives a discrete spectrum and complete basis. The inequality $q_a[u] > a^2\|u\|^2$ for nonzero u gives the strict lower bound.

Set $z = ae^x$, $\nu^2 = -E$. The equation becomes $z^2 y_{zz} + zy_z - (z^2 + \nu^2)y = 0$. Its unique square-integrable solution at infinity is $K_\nu(ae^x)$: it decays as $e^{-ae^x}/\sqrt{ae^x}$, while an independent I solution grows. These standard Bessel identities and conventions are recorded in DLMF [2]. Since K_ν is entire and even in ν, $y_E(x) = K_{\sqrt{-E}}(ae^x)$ is entire in E and independent of the square-root branch. The boundary equation is thus necessary and sufficient for an eigenvalue, including for complex E; self-adjointness excludes nonreal zeros. Uniqueness of the decaying solution proves simplicity. □

Put

$$f_E(x) = K_\nu(a)I_\nu(ae^x) - I_\nu(a)K_\nu(ae^x).$$

The Wronskian $I_\nu' K_\nu - K_\nu' I_\nu = 1/z$ gives $f_E(0) = 0, f_E'(0) = 1$. For $E \notin \sigma(H_a)$ the full kernel is

$$(H_a - E)^{-1}(x, y) = \frac{f_E(\min(x, y))K_\nu(ae^{\max(x,y)})}{K_\nu(a)}. \tag{3}$$

It has derivative jump -1, the Dirichlet boundary and the decaying tail. These verify the inverse first on compactly supported functions, then by the self-adjoint resolvent bound. Any apparent order-branch ambiguity in f_E cancels by uniqueness of its initial-value problem.

Differentiating the equation gives $(H_a - E)\partial_E y_E = y_E$. Therefore the Wronskian $W = y_E(\partial_E y_E)' - y_E'\partial_E y_E$ satisfies $W' = -y_E^2$ and $W(\infty) = 0$. At a real eigenvalue,

$$\|y_n\|^2 = -aK'_{i\sqrt{E_n}}(a)\ \partial_E K_{\sqrt{-E}}(a)\Big|_{E=E_n} > 0. \tag{4}$$

Here prime is argument differentiation. The argument derivative is nonzero by ODE uniqueness, so the energy zero is simple as well. Equation (4) normalizes every member of the complete basis, not just a finite collection of numerically bracketed roots.

3 Degeneration, verification and scope

At $a = 0$ the operator is the free Dirichlet half-line Laplacian, with continuous spectrum $[0, \infty)$ and no compact-resolvent inverse determinant of this convention. As $a \downarrow 0$ the forms decrease on their common core; closure gives strong-resolvent convergence to that free operator. This is a singular confinement boundary, not a uniform limit of the counting constants.

The exact evidence checks 108 complex rational series terms, sixteen action substitutions and twelve derivative/tail majorants. Independent 70-digit calculations check Bessel representations, the differential equation, twelve bracketed source roots, three norm identities and the full resolvent trace against the logarithmic determinant derivative. Numerical calculations are regression observations, not interval certification and not the proof of full spectral completeness. No target zeros or prime table enter the definition or computations.

The source has a natural Friedrichs quantization and complex-conjugation time reversal, but no intrinsic rational-prime carrier. A0 and A1 fail; the target A2/A3 bridge fails, despite a true source logarithmic law. A4 records only natural source quantization. Overall the target candidate is rejected, with all nine target/Route-B flags false and no Route B: `NO_BAD_EULER_OR_ROOT_NUMBER`.

Revision record. Round zero establishes the full Bessel spectrum and resolvent.

References

[1] J. C. Lagarias, The Schrödinger operator with Morse potential on the right half line, *Commun. Number Theory Phys.* 3 (2009), 323–361. doi:10.4310/CNTP.2009.v3.n2.a3; arXiv:0712.3238.

[2] NIST Digital Library of Mathematical Functions, §10.25, §10.27, §10.28, §10.32, §5.11, accessed 5 September 2026.

[3] A. Dobner, Large deviations of the argument of the Riemann zeta function, *Mathematika* 70 (2024), e12251. doi:10.1112/mtk.12251; arXiv:2101.01747v2.